

In Situ Simulation: Suspected COVID-19 (Ambulatory)

Section 1: Case Summary

Scenario Title:	Suspected COVID-19 (Ambulatory)
Keywords:	PPE, traveler, infection control, COVID-19
Brief Description of Case:	A 38 year old male has returned from a business trip in Asia last week. Over the last 48h, he has developed fever, rigors, myalgias as well as nausea, vomiting and upper respiratory symptoms. He initially was trying to ride it out at home, but is feeling increasingly short of breath and fatigued. He has now presented to your ambulatory care clinic. The goal of the case will be to recognize the need to isolate the patient and use appropriate PPE, while assessing and managing his viral illness. The case starts in the triage office, moves to a room with MD assessment and may also involve handover to EMS for transfer.

Goals and Objectives	
Educational Goal:	Examine the processes around identification of infectious presentations requiring isolation and advanced infection control procedures, as well as use of PPE
Objectives: (Medical and CRM)	1. Early recognition of infectious symptoms in a returning traveler and the need for infection control procedures 2. Appropriate use of PPE, including N95 masks and donning and doffing procedures, as detailed in local guidelines 3. Use of a negative pressure room or alternative when available
EPAs Assessed:	N/A

Learners, Setting and Personnel			
Target Learners:	☐ Junior Learners	☐ Senior Learners	☒ Staff
	☒ Physicians	☒ Nurses	☐ RTs
	☒ Other Learners:		
Location:	☐ Sim Lab	☒ In Situ	☐ Other:
Recommended Number of Facilitators:	Instructors: facilitators to observe		
	Confederates: None		
	Sim Techs: None		

Scenario Development	
Date of Development:	February 10 th , 2020
Scenario Developer(s):	Dr. Alex Chorley, MD
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Last Revision Date:	
Revised By:	
Version Number:	1



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Section 2A: Initial Patient Information

A. Patient Chart					
Patient Name: Tony Stark		Age: 38		Gender: M	Weight: 70kg
Presenting complaint: Shortness of Breath					
Temp: 39.1	HR: 120	BP: 130/75	RR: 25	O ₂ Sat: 89%	FiO ₂ : R/A
Cap glucose: 5.2			GCS: 15		
Triage note: The patient has returned from a business trip in Wuhan, China last week. Over the last 48h, he has developed fever, rigors, myalgias as well as nausea, vomiting and upper respiratory symptoms. He initially was trying to ride it out at home, but is feeling increasingly short of breath and fatigued. Hasn't taken any antipyretics today. No diarrhea.					
Allergies: Sulfa					
Past Medical History: None			Current Medications: None		

Section 2B: Extra Patient Information

A. Further History	
<i>Include any relevant history not included in triage note above. What information will only be given to learners if they ask? Who will provide this information (mannequin's voice, confederate, SP, etc.)?</i>	
N/A	

B. Physical Exam	
<i>List any pertinent positive and negative findings</i>	
Cardio: Normal HS	Neuro: Alert and oriented, no focal deficits
Resp: Tachypneic, mild WOB, crackles to RLL	Head & Neck: Normal but congested
Abdo: Soft non tender	MSK/skin: no rash, cap refill <2s
Other:	

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Section 3: Technical Requirements/Room Vision

A. Patient
☐ Mannequin (<i>specify type and whether infant/child/adult</i>)
☒ Standardized Patient
☐ Task Trainer
☐ Hybrid
B. Special Equipment Required
PPE: N95 Masks, gowns, gloves, face shield. Negative pressure room if available.
C. Required Medications
None.
D. Moulage
None.
E. Monitors at Case Onset
☐ Patient on monitor with vitals displayed
☒ Patient not yet on monitor
F. Patient Reactions and Exam
<i>Include any relevant physical exam findings that require mannequin programming or cues from patient (e.g. – abnormal breath sounds, moaning when RUQ palpated, etc.) May be helpful to frame in ABCDE format.</i>
As above

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Section 4: Confederates and Standardized Patients

Confederate and Standardized Patient Roles and Scripts	
Role	Description of role, expected behavior, and key moments to intervene/prompt learners. Include any script required (including conveying patient information if patient is unable)
Patient	You returned 5 days ago from a 1 week trip to China, including Wuhan. You felt fine on the plane and for the first few days, then developed cough, runny nose and sore throat. You went on to develop fever, chills, shakes and muscle aches. There have been several episodes of nausea and vomiting but no diarrhea. There hasn't been any blood in your vomit. You feel like it's been increasingly hard to breathe and you are breathing more quickly than usual. The cough is bringing up some green phlegm.

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Section 5: Scenario Progression

Scenario States, Modifiers and Triggers				
Patient State/Vitals	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State		Facilitator Notes
1. Triage Office Rhythm: Sinus Tach HR: 120 BP: 130/75 RR: 25 O ₂ SAT: 89% T: 39.1°C GCS: 15	<i>Patient has mild tachypnea and work of breathing</i>	<u>Expected Learner Actions</u> ☐ Triage patient (brief Hx&Px) ☐ Respiratory illness screening ☐ Perform vitals ☐ Have patient wear a mask ☐ Notify charge nurse of need for isolation	<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> <u>Triggers</u> <i>For progression to next state</i> Patient moved to room → 2. Isolation Room	
2. Isolation Room Rhythm: Sinus Tach HR: 120 BP: 130/75 RR: 25 O ₂ SAT: 89% T: 39.1°C GCS: 15	<i>Unchanged</i>	<u>Expected Learner Actions</u> ☐ Move patient to negative pressure room or alternative ☐ Don appropriate PPE (Per local protocols) ☐ Brief History & Physical ☐ Notify MD, contact Infection Control ☐ Apply O ₂ ☐ Doff PPE on exit of room	<u>Modifiers</u> O ₂ applied → SpO ₂ to 95% <u>Triggers</u> MD arrival → 3. MD Assessment	
3. MD Assessment Rhythm: Sinus Tach HR: 120 BP: 130/75 RR: 25 O ₂ SAT: 89% T: 39.1°C GCS: 15	<i>Unchanged</i>	<u>Expected Learner Actions</u> ☐ MD dons appropriate PPE ☐ History and Physical ☐ CXR (Portable) ☐ Bloodwork (incl Cultures) ☐ IV Fluids ☐ Antibiotics (Ceftriaxone + Azithro vs Fluroquinolone) ☐ Contact ID re: swabs/work up	<u>Modifiers</u> IV Fluid bolus → HR to 105 O ₂ applied → SpO ₂ to 95% <u>Triggers</u> ID/Infection control contacted → End Case -OR- EMS contacted → 4. Transport	



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4. Transport (optional) Rhythm: Sinus Tach HR: 105 BP: 130/75 RR: 25 O ₂ SAT: 95% (NP) T: 39.1°C GCS: 15		<u>Expected Learner Actions</u> ☐ Pt PPE for transport ☐ PPE for EMS ☐ Clinical handover to EMS ☐ Arrange appropriate cleaning of room and equipment	<u>Modifiers</u> - - - <u>Triggers</u> - Handover to EMS Team → End Case	
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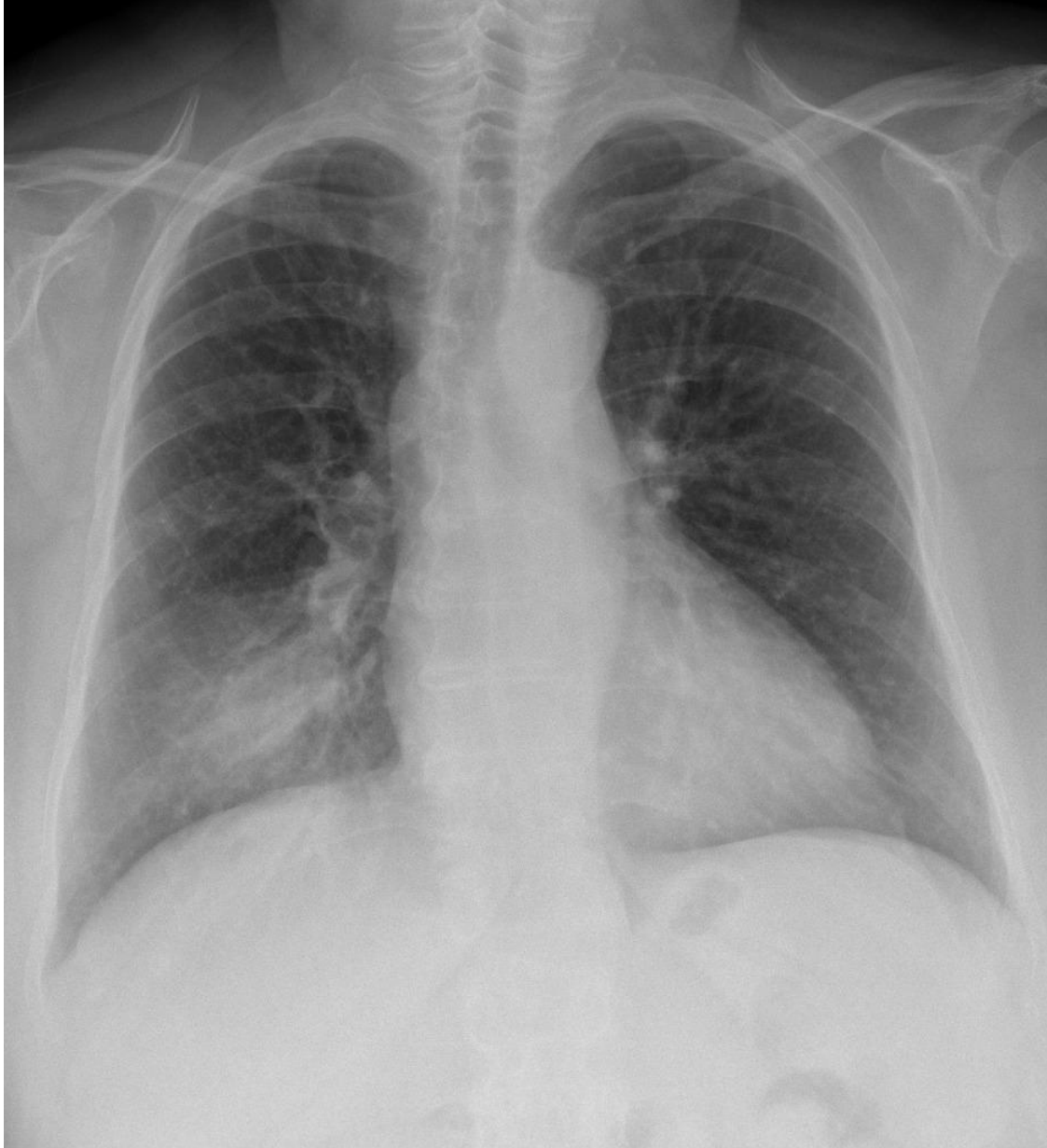
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Appendix A: Laboratory Results

None given during scenario

Appendix B: ECGs, X-rays, Ultrasounds and Pictures

Paste in any auxiliary files required for running the session. Don't forget to include their source so you can find them later!



Case courtesy of Dr Henry Knipe, [Radiopaedia.org](https://radiopaedia.org), rID: 31352

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Appendix C: Facilitator Cheat Sheet & Debriefing Tips

Include key errors to watch for and common challenges with the case. List issues expected to be part of the debriefing discussion. Supplemental information regarding any relevant pathophysiology, guidelines, or management information that may be reviewed during debriefing should be provided for facilitators to have as a reference.

In Situ Simulation for Emerging Diseases

In situ simulation is a team-based training technique conducted in actual patient care areas using equipment and supplies from that area with people from the care team. In situ simulation has been used successfully to identify latent safety threats, including fixable problems with teamwork, systems issues and adherence to guidelines.

In the case of COVID-19 and other novel infectious diseases, guidelines are distributed that often present a challenge for frontline staff because they are both different from usual procedures and rapidly evolving. In situ simulation offers an opportunity to practice using these guidelines and identify issues that are preventing staff from adhering to these guidelines.

This case should be run with local guidelines in mind and may include a diverse group of stakeholders (front-line workers, infection prevention and control, infectious disease specialists and administrators).

Some Questions to Explore during the Debrief

- How do we use PPE to keep providers safe? How do we effectively use the PPE?
- How do we keep other patients safe?
- What space will we use to care for the patient? How is this cleaned?
- What are some high risk activities to be aware of?
- What are the policies on bringing equipment in and out of the room?
- Who needs to be notified/consulted in this case?
- How do we coordinate transfer/handover to other services and providers?
- How do we find the most up-to-date policies and procedures?

References

1. <https://www.who.int/emergencies/diseases/novel-coronavirus-2019>
2. <https://www.cdc.gov/coronavirus/index.html>
3. <https://ipac-canada.org/coronavirus-resources.php>

